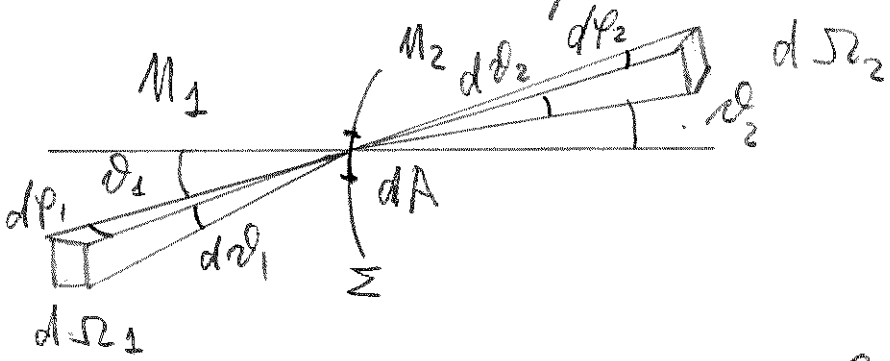
Interface between two optical media



$$d\Omega_1 = \sin\vartheta_1 \, d\vartheta_1 \, d\varphi_1$$

$$d\Omega_2 = \sin\vartheta_2 \, d\vartheta_2 \, d\varphi_2$$

Snell' law

$$n_1 \sin\vartheta_1 = n_2 \sin\vartheta_2$$

$$n_1 \cos\vartheta_1 \, d\vartheta_1 = n_2 \cos\vartheta_2 \, d\vartheta_2$$

$$d\varphi_1 = d\varphi_2$$

$$\frac{d\Omega_1}{d\Omega_2} = \frac{\sin\vartheta_1 \, d\vartheta_1}{\sin\vartheta_2 \, d\vartheta_2}$$

$$\frac{d\Omega_1}{d\Omega_2} = \frac{n_2}{n_1} \frac{n_2}{n_1} \frac{\cos\vartheta_2}{\cos\vartheta_1}$$

$$\frac{d\Omega_1}{d\Omega_2} = \frac{n_2^2}{n_1^2} \frac{\cos\vartheta_2}{\cos\vartheta_1}$$

$$d\Phi = L_1 \, dA \cos\vartheta_1 \, d\Omega_1 = L_2 \, dA \cos\vartheta_2 \, d\Omega_2$$

$$L_2 = \frac{L_1 \, dA \cos\vartheta_1 \, d\Omega_1}{dA \cos\vartheta_2 \, d\Omega_2} = L_1 \frac{\cos\vartheta_1}{\cos\vartheta_2} \frac{n_2^2}{n_1^2} \frac{\cos\vartheta_2}{\cos\vartheta_1}$$

$$\frac{L_2}{n_2^2} = \frac{L_1}{n_1^2}$$ ← Basic radiance

The basic radiance is preserved under refraction
→ the basic radiance is invariant in whatsoever optical system.

$$\frac{L_1}{n_1^2} = \frac{L_2}{n_2^2} = \dots \frac{L_n}{n_n^2}$$

Photon losses due to scattering and absorption modify the basic radiance
O.L apertures do not even if they reduce the radiant flux

POLITECNICO DI MILANO POLITECNICO DI MILANO POLITECNICO DI MILANO POLITECNICO DI MILANO POLITECNICO DI MILANO POLITECNICO DI MILANO

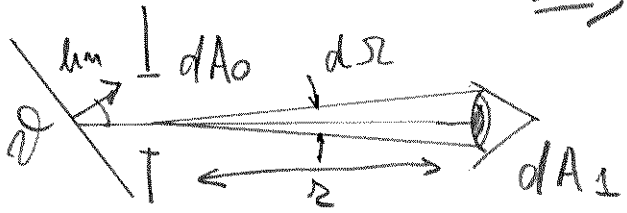
Lambertian sources

Ligh sources, either primary or secondary, that appear alike when observed from any angle are called Lambertian and show the same property

$$L(x,y;\vartheta,\varphi) \rightarrow L(x,y) \qquad \text{or, for small sources } L(x,y;\vartheta,\varphi) = L_0$$

Lambertian source → no change when changing the abs. angle ⟹

$$\Longrightarrow L(\vartheta,\varphi) = L_0$$



The flux emitted by the source and received by the detector is:

$$d\phi = L(\vartheta,\varphi)\, dA_\perp\, d\Omega = L(\vartheta,\varphi)\, dA_\perp \frac{dA_1}{r^2}$$

$$dA_\perp = dA_0 \quad \forall \vartheta \rightarrow d\phi = L(\vartheta,\varphi)\frac{dA_0\, dA_1}{r^2}$$

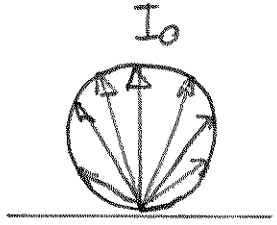
The requirement that $d\phi$ does not depend on $\vartheta \rightarrow L = L(\varphi)$

Rotating the source by $90^\circ \rightarrow L(\varphi) = L_0 \Longrightarrow L(\vartheta,\varphi) = L_0$

Radiant intensity of a Lambertian source

$$I(\vartheta,\varphi) = \frac{d\phi}{d\Omega} = \int_{A_0} \frac{L_0\, dA \cos\vartheta\, \cancel{d\Omega}}{\cancel{d\Omega}} = L_0 A_0 \cos\vartheta$$

$$I(\vartheta) = I_0 \cos\vartheta = L\,[A_0 \cos\vartheta]$$



The intensity of a Lambertian source scales with the apparent area

POLITECNICO DI MILANO POLITECNICO DI MILANO POLITECNICO DI MILANO POLITECNICO DI MILANO POLITECNICO DI MILANO POLITEC

Radiant flux emitted by a Lambertian source

$$\phi = \int_{2\pi} I_0 \cos\vartheta \, d\Omega = I_0 \int_0^{2\pi} d\varphi \int_0^{\pi/2} \cos\vartheta \sin\vartheta \, d\vartheta = \pi I_0$$

$$\Rightarrow \quad I_0 = \phi/\pi \qquad \text{Isotropic source} \qquad \underbrace{I_0 = \phi/4\pi}_{4\pi} \qquad \underbrace{I_0 = \phi_{1/2}/2\pi}_{2\pi}$$

Exitance of a Lambertian source

$$M(x,y) = \frac{d\phi}{dA} \int_{2\pi} \frac{L(x,y)\, \cancel{dA} \cos\vartheta \, d\Omega}{\cancel{dA}} = L(x,y)\, 2\pi \int_0^{\pi/2} \cos\vartheta \sin\vartheta \, d\vartheta = L(x,y)\pi$$

Uniform source $\rightarrow M_0 = L_0 \pi$

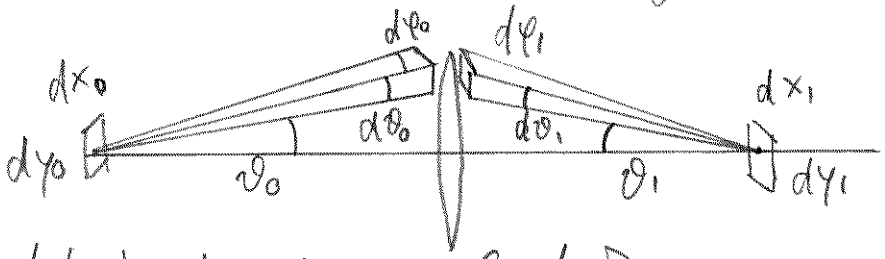
Summary

$$\phi = I_0 \pi = L_0 A_0 \pi = M_0 A_0$$

ϕ is known	L_0 is known
$I_0 = \phi/\pi$	$I_0 = L_0 A_0$
$L_0 = \dfrac{\phi}{\pi A_0}$	$\phi = L_0 A_0 \pi$
$M_0 = \dfrac{\phi}{A_0}$	$M_0 = L_0 \pi$

POLITECNICO DI MILANO POLITECNICO DI MILANO POLITECNICO DI MILANO POLITECNICO DI MILANO POLITECNICO DI MILANO POLITECNICO DI MILANO POLITECI

Radiance of an image made by an aplanatic system



Abbe's condition

$$n_0\, dx_0 \sin\vartheta_0 = n_1\, dx_1 \sin\vartheta_1$$

$$n_0\, dy_0 \cos\vartheta_0\, d\vartheta_0 = n_1\, dy_1 \cos\vartheta_1\, d\vartheta_1$$

$$n_0^2\, dx_0\, dy_0 \sin\vartheta_0 \cos\vartheta_0\, d\vartheta_0 = n_1^2\, dy_1\, dy_1 \sin\vartheta_1 \cos\vartheta_1\, d\vartheta_1$$

$$d\Phi = L_0\, dx_0\, dy_0 \cos\vartheta_0\, d\Omega_0$$

$$d\Phi = L_0\, \underbrace{dx_0\, dy_0 \cos\vartheta_0 \sin\vartheta_0\, d\vartheta_0}\, d\varphi$$

$$d\Phi = L_0 \frac{n_1^2}{n_0^2}\, dx_1\, dy_1 \cos\vartheta_1 \sin\vartheta_1\, d\vartheta_1\, d\varphi$$

$$d\Phi = L_0 \frac{n_1^2}{n_0^2}\, dA_1 \cos\vartheta_1\, d\Omega_1$$

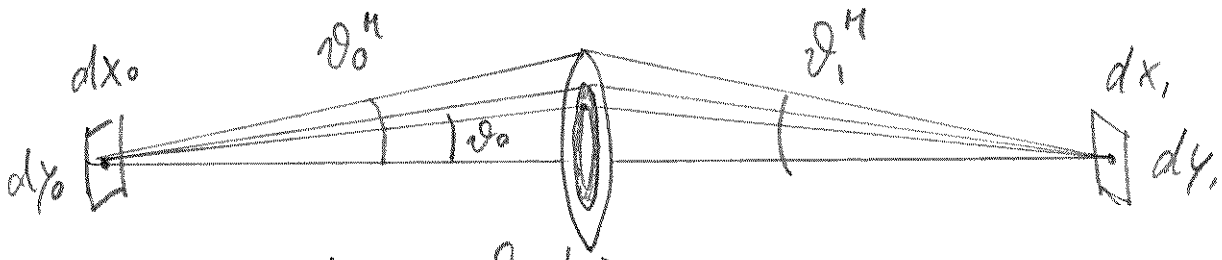
$$\left[\frac{d\Phi}{dA_1 \cos\vartheta_1\, d\Omega_1} = L_0 \frac{n_1^2}{n_0^2}\right]$$

$$L_1 = \frac{d\Phi}{dA_1 \cos\vartheta_1\, d\Omega_1} = L_0 \frac{n_1^2}{n_0^2}$$

$$\frac{L_1}{n_1^2} = \frac{L_0}{n_0^2}$$

POLITECNICO DI MILANO POLITECNICO DI MILANO POLITECNICO DI MILANO POLITECNICO DI MILANO POLITECNICO DI MILANO POLITEC

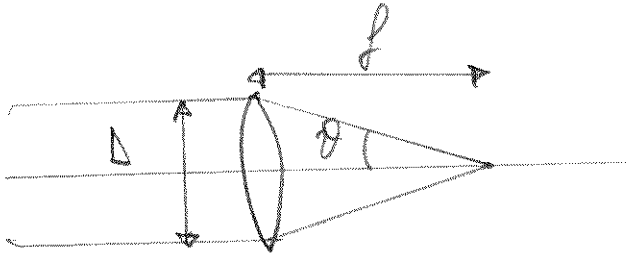
Irradiance of the image of a Lambertian source made by an aplanatic system



Abbe's condition

$$n_0\, dx_0 \sin\vartheta_0 = n_1\, dx_1 \sin\vartheta_1$$
$$n_0\, dy_0 \sin\vartheta_0 = n_1\, dx_1 \sin\vartheta_1$$
$$n_0^2\, \underline{dx_0\, dy_0 \sin^2\vartheta_0} = n_1^2\, \underline{dx_1\, dy_1 \sin^2\vartheta_1}$$

$$d\phi = L_0\, dx_0\, dy_0 \cos\vartheta_0\, d\Omega_0$$
$$d\phi = L_0\, dx_0\, dy_0 \cos\vartheta_0 \sin\vartheta_0\, d\vartheta_0\, d\varphi$$
$$d\phi = L_0\, dx_0\, dy_0 \int_0^{2\pi} d\varphi \int_0^{\vartheta_0^M} \sin\vartheta_0 \cos\vartheta_0\, d\vartheta_0$$
$$d\phi = L_0\, dx_0\, dy_0\, \cancel{2}\pi\, \frac{\sin^2\vartheta_0^M}{\cancel{2}}$$
$$d\phi = \pi L_0 \frac{n_1^2}{n_0^2}\, dx_1\, dy_1 \sin^2\vartheta_1^M$$



$$f/\# = f/D = \frac{1}{2\tan\vartheta} \simeq \frac{1}{2\sin\vartheta}$$

$$E_1 = \frac{d\phi}{dA_1} = \pi L_0 \frac{n_1^2}{n_0^2} \sin^2\vartheta_1^M$$

$$E_1 = \pi L_0 \frac{n_1^2}{n_0^2} \frac{1}{4} \frac{1}{f/\#^2}$$

Example

$$\left[f/\# = f/2 \xrightarrow[n_1 = n_0]{} E = \frac{1}{16} M \right]$$

$$E_1 = M_0 \frac{n_1^2}{n_0^2} \frac{1}{4\, f/\#^2} \qquad \boxed{M_L \ll 1}$$

POLITECNICO DI MILANO POLITECNICO DI MILANO POLITECNICO DI MILANO